

Amarnath Nanduri

Electrical & Electronics Engineer

✉ amarnathnanduri008@gmail.com ☎ 010-5927-8152 🔗 nanduri-amrnath.github.io

PROFILE

Motivated Electrical & Electronics Engineering student focused on embedded systems, IoT, and power electronics. Seeking internships or research opportunities to apply hardware prototyping and signal processing skills.

EDUCATION

Kwangwoon University, Seoul , Semester Exchange	Present
KCG College Of Technology, Anna University , Bachelor of Engineering, Electrical & Electronics	Present

SKILLS

Technical — Programming: C, C++, Python, MATLAB | Hardware Platforms: Raspberry Pi, ESP32, Arduino Tools: MATLAB/Simulink, Multisim, Proteus, AutoCAD Electrical | Core Domains: Power Electronics, Smart Grid, VLSI, Signal Process, **Non-Technical** — | Teamwork, Communication, Leadership, Time Management, Problem Solving, Adaptability, Project Management, Analytical Thinking.

PROJECTS

AcaDexa v2 — Academic Management Platform, Smart Schedule(collaborative) 🔗

IoT Elephant Redirection System, device with ultrasonic redirection to prevent elephant–train collision.

Smart Tire Monitoring System, Multi-sensor tire health system measuring with real time alerting.

AI-Based Rail Defect Detection, used ML classifiers to detect cracks based on frequency-domain signatures.

Adaptive Low Power ALU, Power-efficient 8-bit ALU with dynamic module-level clock gating.

Bidirectional Converter, predictive bidirectional DC–DC converter.

Detailed Description(GitHub)

RESEARCH INTERESTS

Embedded Systems | IoT for Smart Infrastructure | Sustainable Power and Energy Systems.
| Fault Detection in Electrical Networks. | AI/ML